SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**M. Kyathi Sai Priya**
Register Number	**192572101**

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications.* (BL3)
Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3) **Total Marks: 40**
Question Count: 4

Duration : 60 Minutes

Code of Conduct
I _M.Kyathi Sai Priya_ (192572101) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: _M.Kyathi__

Experiments Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 - Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Ans :

Aim:

To create a cloud-based Library Book Reservation System using Zoho Creator.

Algorithm :

1.Open the selected cloud service provider.

2.Create a new cloud application.

3.Create a form named Library Book Reservation.

4.Add fields for book details.

5.Add fields for reservation and availability status.

6.Configure the form and save it.

7.Add sample book records.

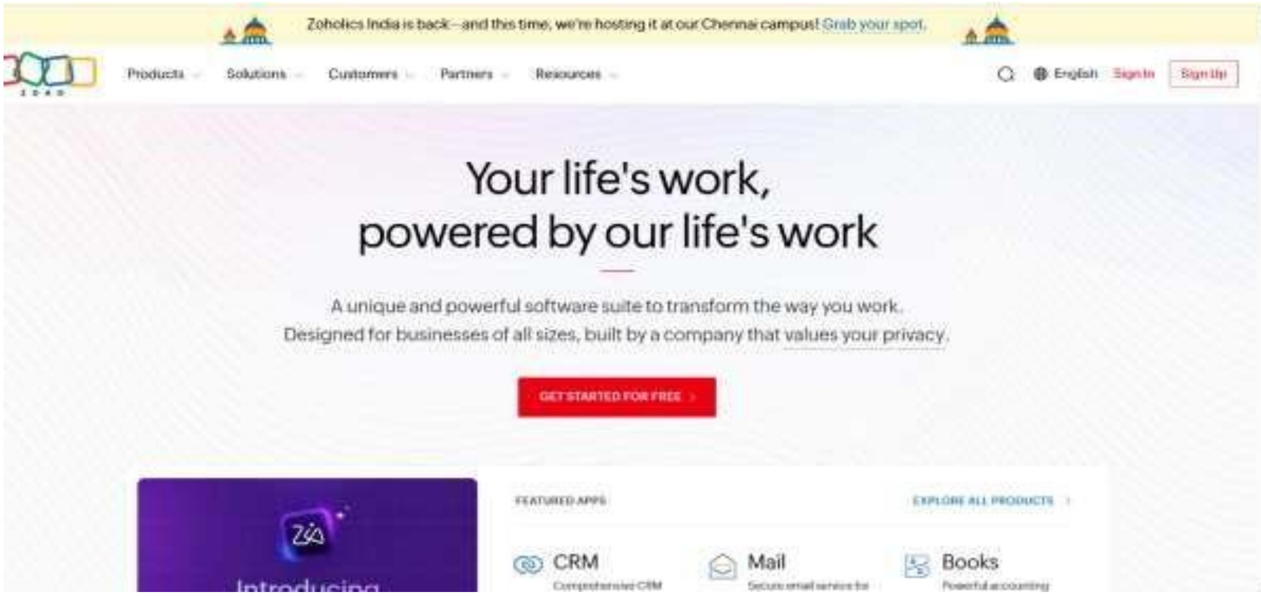
8.Create a report/list to display the books.

9.Test the application by adding and updating book records.

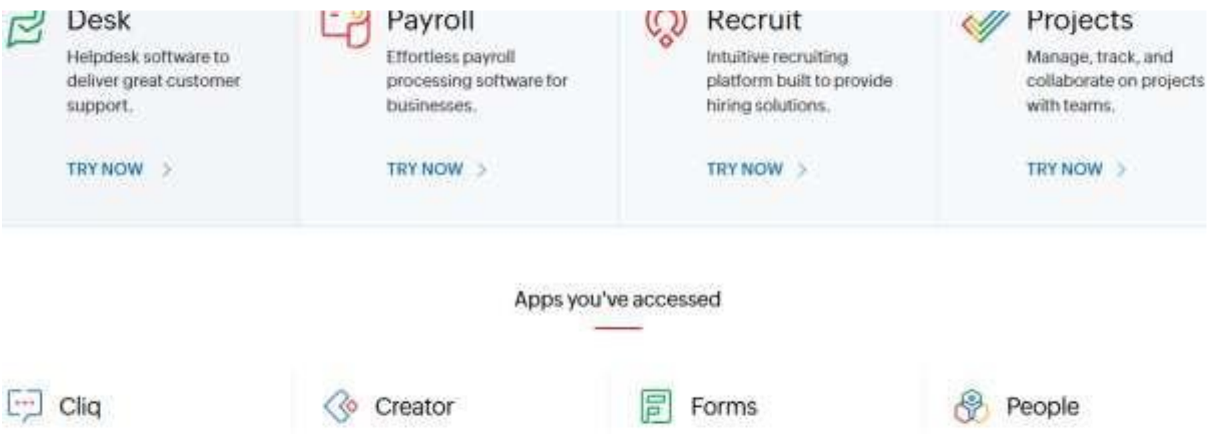
10.Verify that the application works through the cloud.

Implementation :

STEP 1: Go to zoho.com

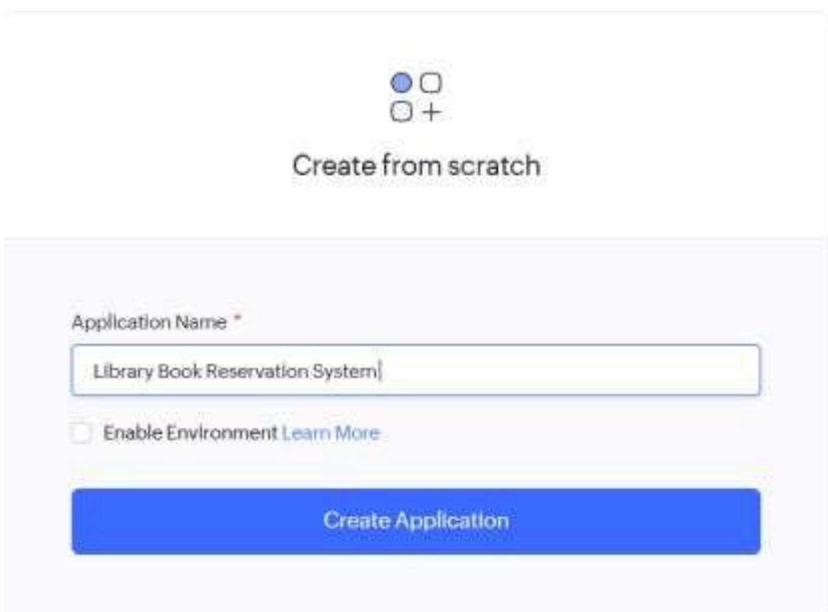


STEP 2: Login to your Zoho account and open Zoho Creator c

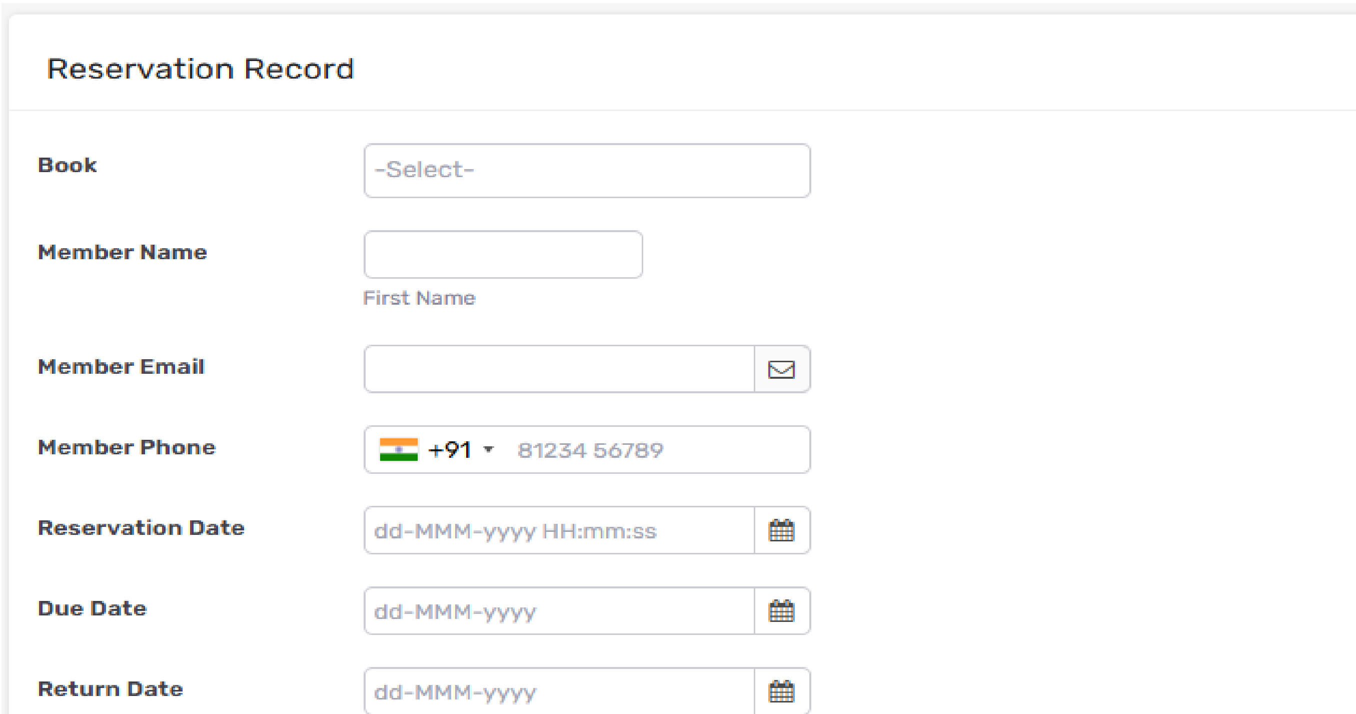


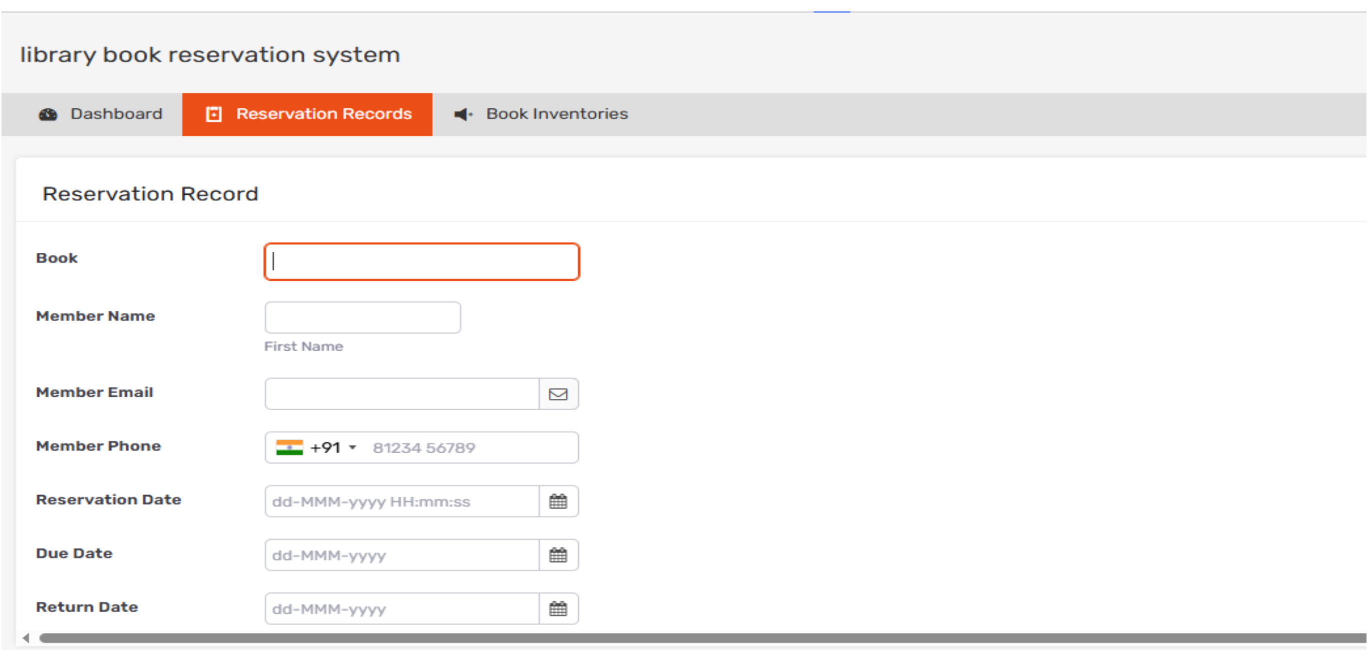
STEP 3: Click Create New Application and select Create from Scratch

STEP 4: Enter Library Book Reservation System as the application name and click Create Application**.**

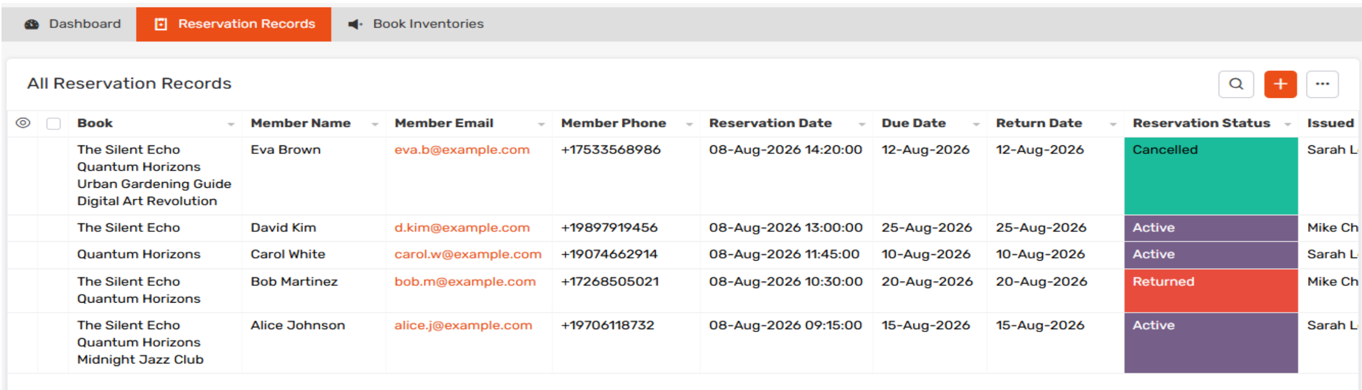


Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

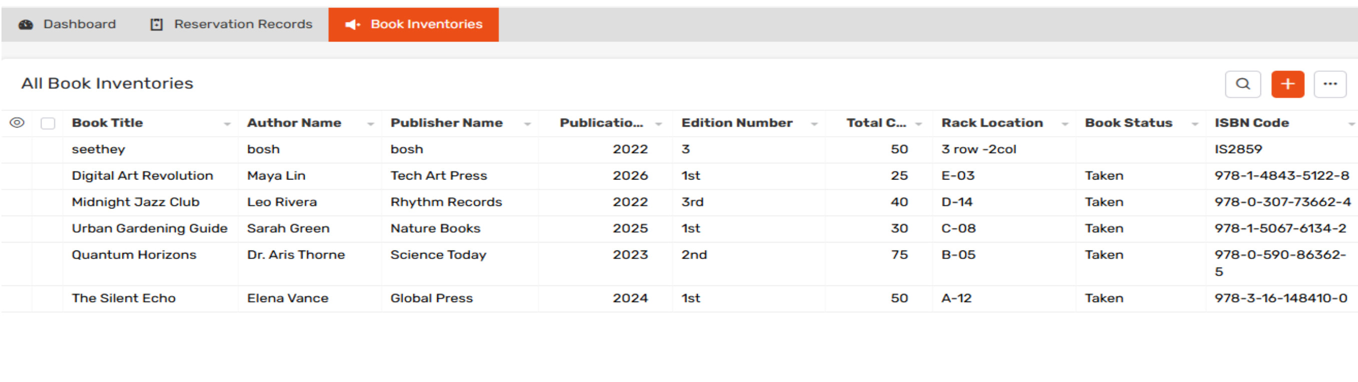

Reservation Record
Book
-Select-
Member Name
First Name
Member Email
Member Phone
+91
81234 56789
Reservation Date
dd-MMM-yyyy HH:mm:ss
Due Date
dd-MMM-yyyy
Return Date
dd-MMM-yyyy


library book reservation system
Dashboard
Reservation Records
Book Inventories
Reservation Record
Book
Member Name
First Name
Member Email
Member Phone
+91
81234 56789
Reservation Date
dd-MMM-yyyy HH:mm:ss
Due Date
dd-MMM-yyyy
Return Date
dd-MMM-yyyy

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

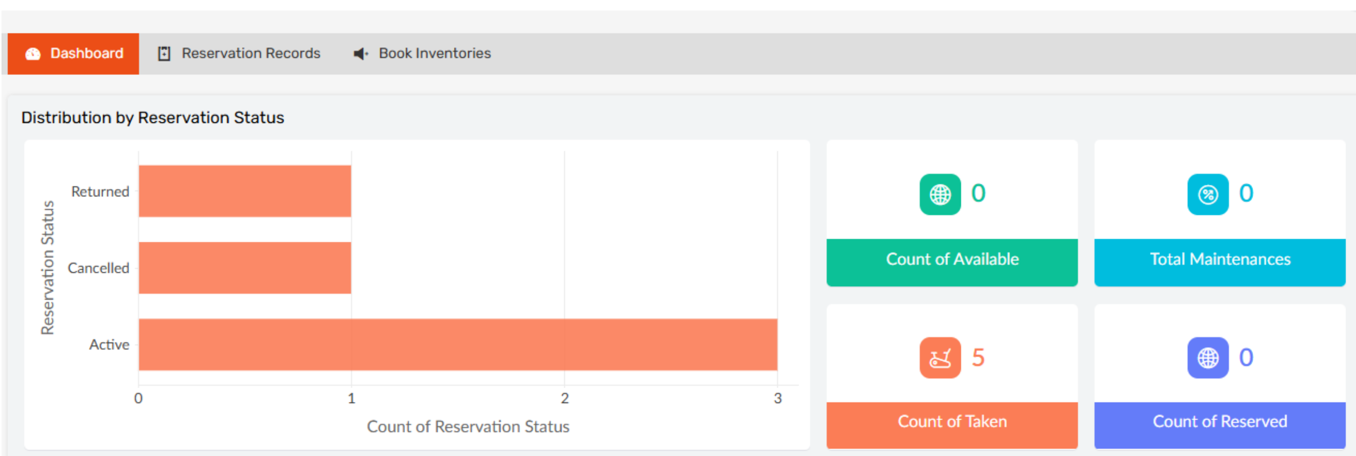


Book	Member Name	Member Email	Member Phone	Reservation Date	Due Date	Return Date	Reservation Status	Issued
The Silent Echo Quantum Horizons Urban Gardening Guide Digital Art Revolution	Eva Brown	eva.b@example.com	+17533568986	08-Aug-2026 14:20:00	12-Aug-2026	12-Aug-2026	Cancelled	Sarah L
The Silent Echo	David Kim	d.kim@example.com	+19897919456	08-Aug-2026 13:00:00	25-Aug-2026	25-Aug-2026	Active	Mike Ch
Quantum Horizons	Carol White	carol.w@example.com	+19074662914	08-Aug-2026 11:45:00	10-Aug-2026	10-Aug-2026	Active	Sarah L
The Silent Echo Quantum Horizons	Bob Martinez	bob.m@example.com	+17268505021	08-Aug-2026 10:30:00	20-Aug-2026	20-Aug-2026	Returned	Mike Ch
The Silent Echo Quantum Horizons Midnight Jazz Club	Alice Johnson	alice.j@example.com	+19706118732	08-Aug-2026 09:15:00	15-Aug-2026	15-Aug-2026	Active	Sarah L



Book Title	Author Name	Publisher Name	Publicatio...	Edition Number	Total C...	Rack Location	Book Status	ISBN Code
seethey	bosh	bosh	2022	3	50	3 row -2col		IS2859
Digital Art Revolution	Maya Lin	Tech Art Press	2026	1st	25	E-03	Taken	978-1-4843-5122-8
Midnight Jazz Club	Leo Rivera	Rhythm Records	2022	3rd	40	D-14	Taken	978-0-307-73662-4
Urban Gardening Guide	Sarah Green	Nature Books	2025	1st	30	C-08	Taken	978-1-5067-6134-2
Quantum Horizons	Dr. Aris Thorne	Science Today	2023	2nd	75	B-05	Taken	978-0-590-86362-5
The Silent Echo	Elena Vance	Global Press	2024	1st	50	A-12	Taken	978-3-16-148410-0

Step 7: The Software has Been Created



Result : Therefore, created a Library Book Reservation System using Zoho Creator including SaaS and also successfully got the results for the library book reservation system

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Ans :

Aim:

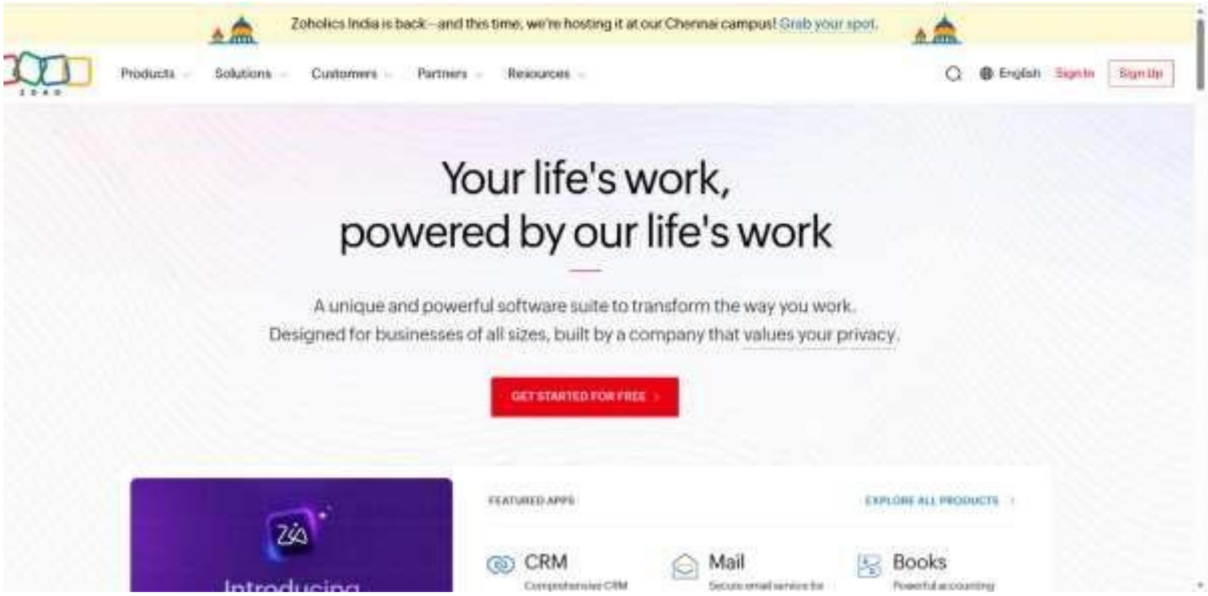
To create a cloud-based Payroll Processing System using Zoho Creator.

Algorithm :

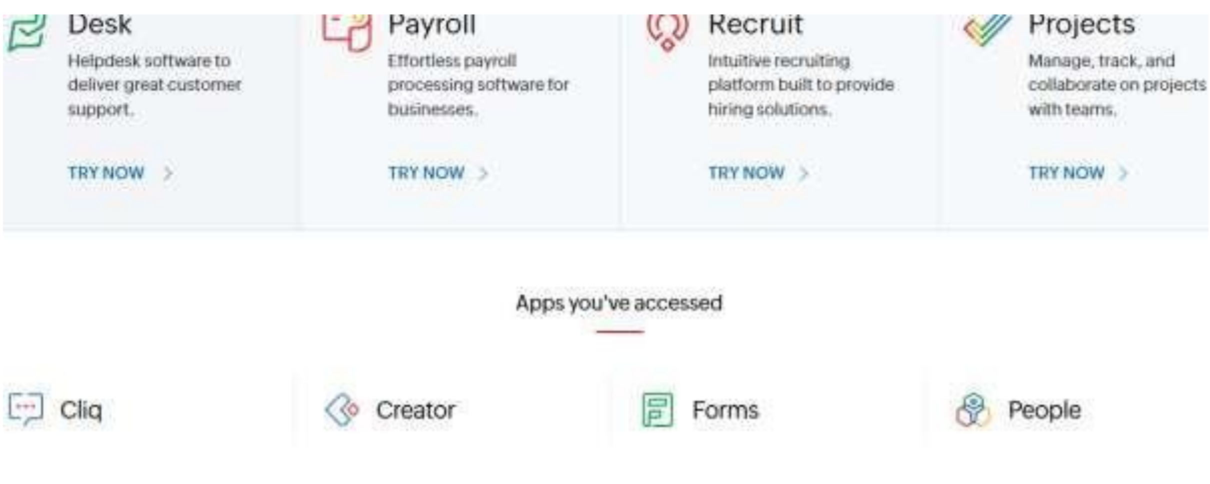
1. Open the selected cloud service provider.
2. Create a new cloud application.
3. Create a form named Employee Payroll.
4. Add employee information fields.
5. Add salary-related fields.
6. Calculate allowances such as HRA, DA, and TA.
7. Calculate gross pay.
8. Calculate net pay.
9. Create a payroll report.
10. Enter sample employee records and verify the salary calculation

.**Implementation :**

STEP 1: Go to zoho.com

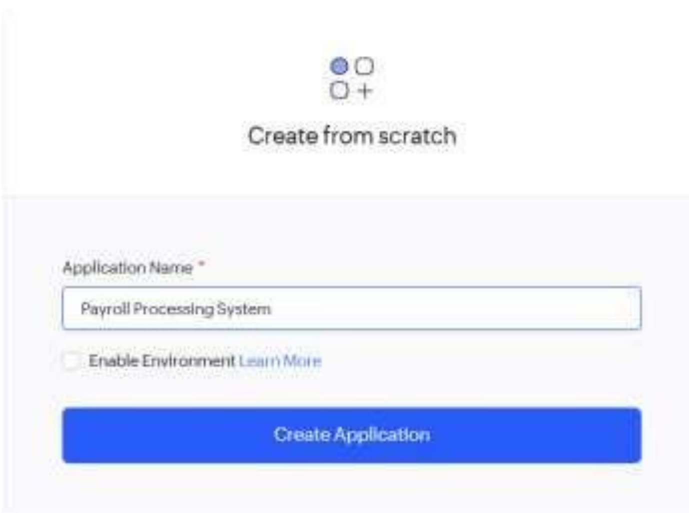


STEP 2: Login to your Zoho account and open Zoho Creator

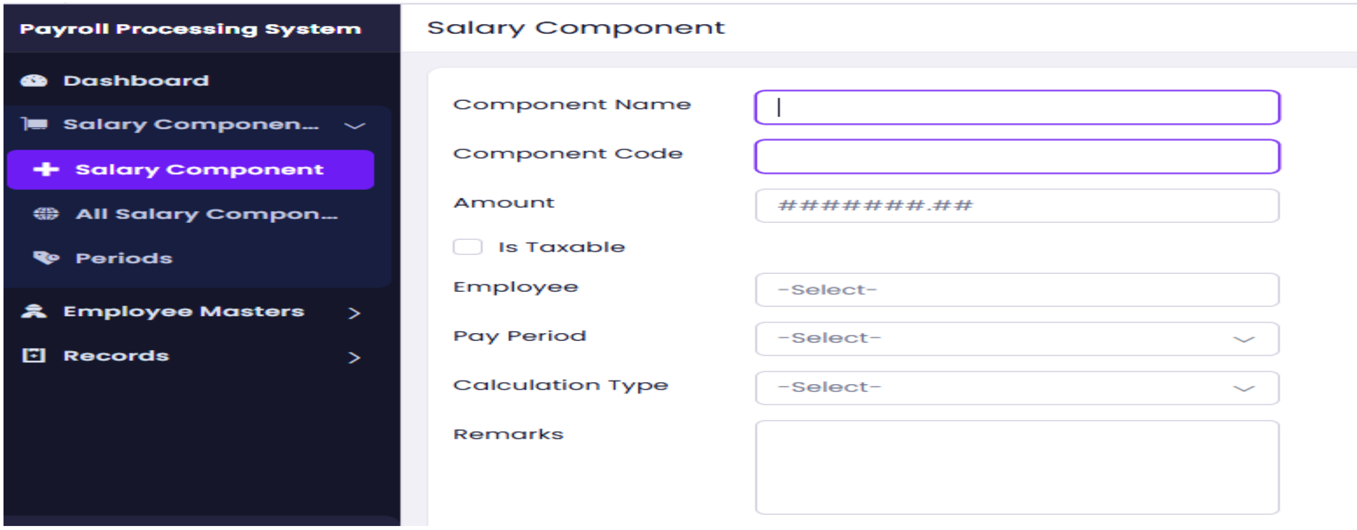


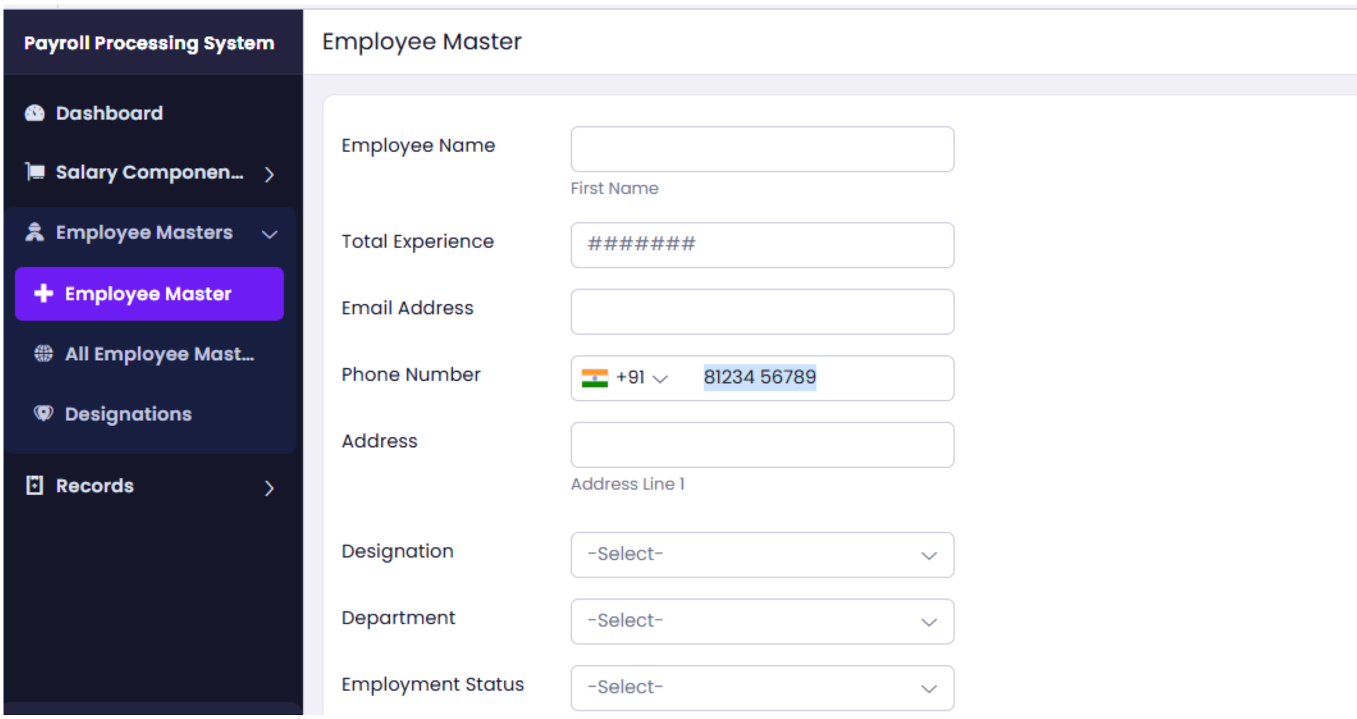
STEP 3: Click Create New Application and select Create from Scratch

STEP 4: Enter Payroll Processing System as the application name and click Create Application**.**

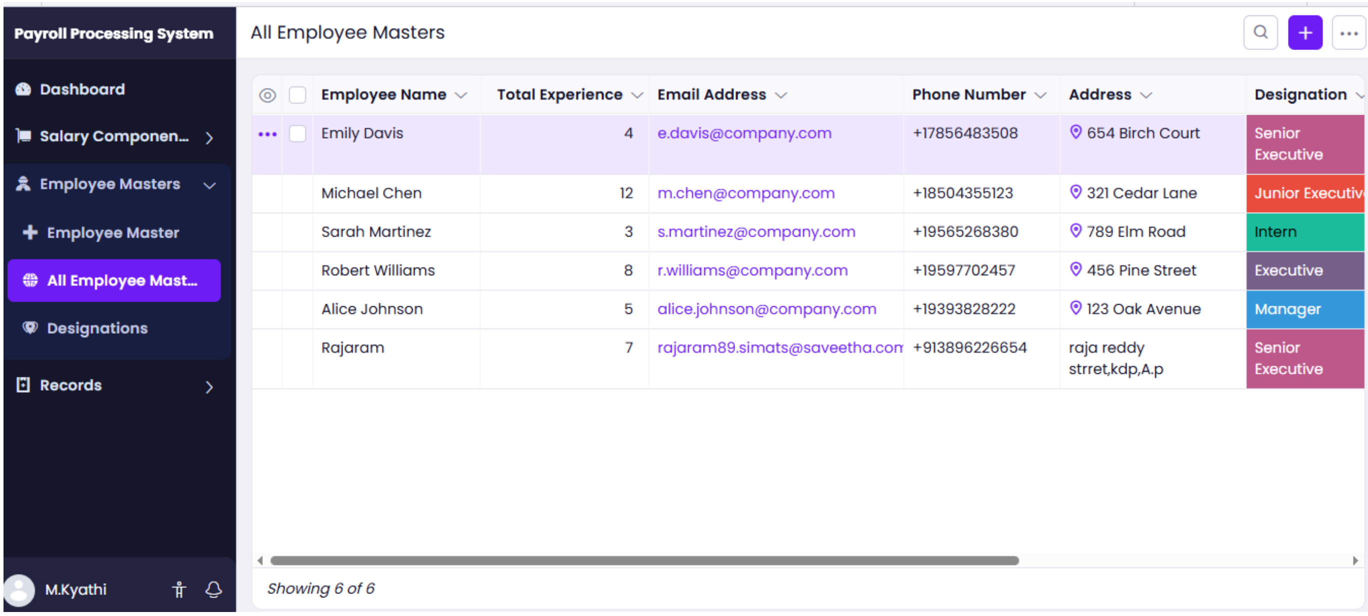


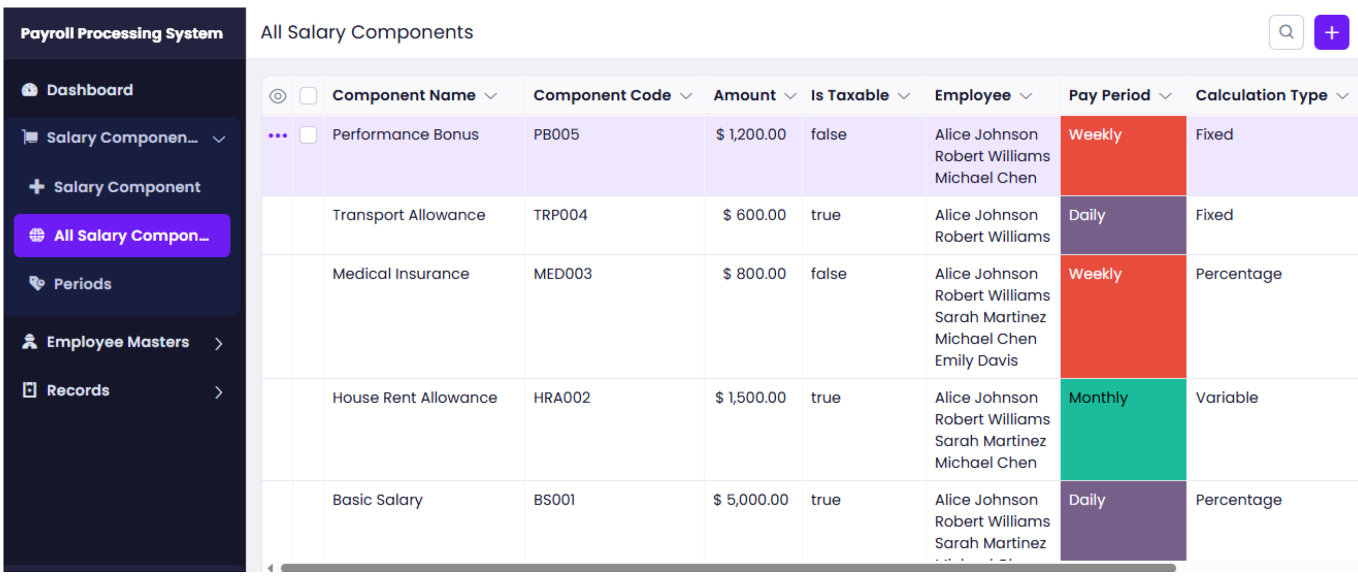
STEP 5 : Add fields names such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

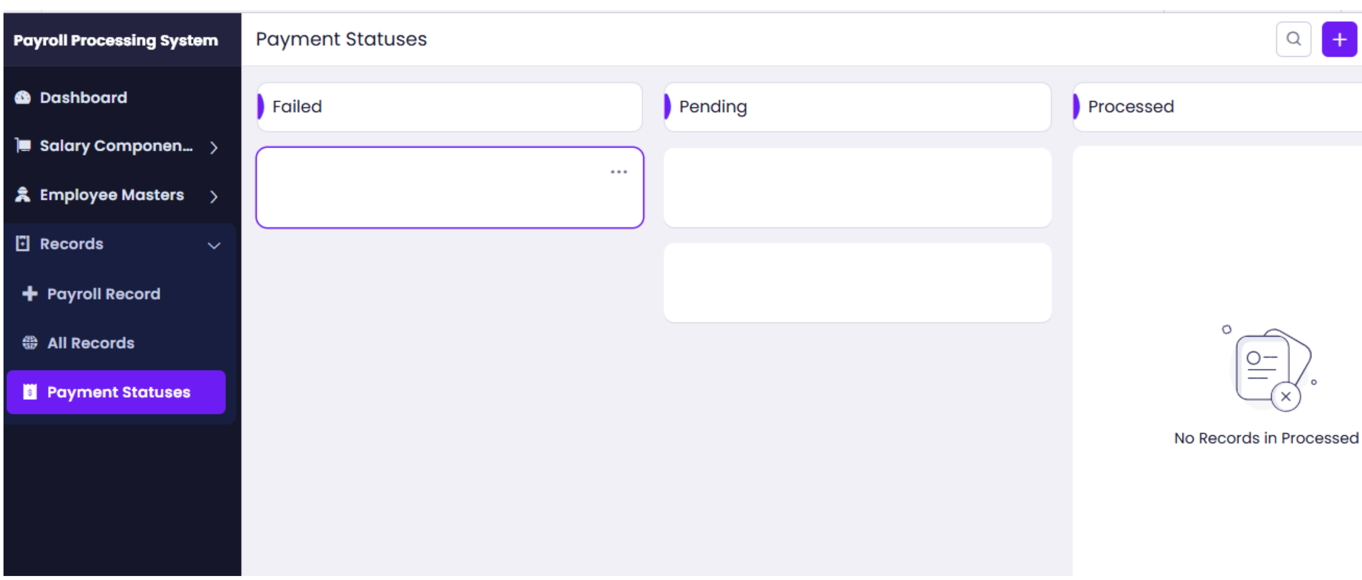




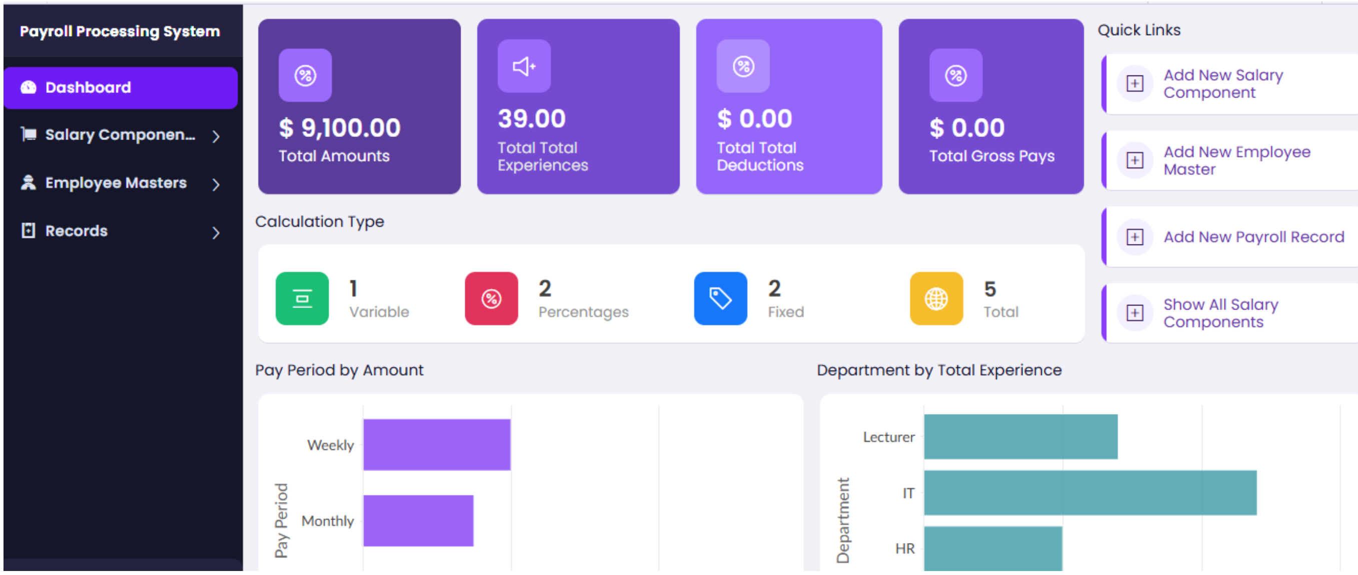
STEP 6: Save the form Payroll Processing, enter sample booking records, and view them in the System.







STEP 7 :The Software has Been Created



Result : Hence, I have created a Pay Roll Processing System using Zoho Creator including SaaS and implemented Successfully

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Ans :

Aim

To demonstrate virtualization by installing a Type-2 Hypervisor (Oracle VirtualBox) and creating a Virtual Machine with 1 CPU, 2 GB RAM, and 15 GB storage using a Windows/Linux host operating system.

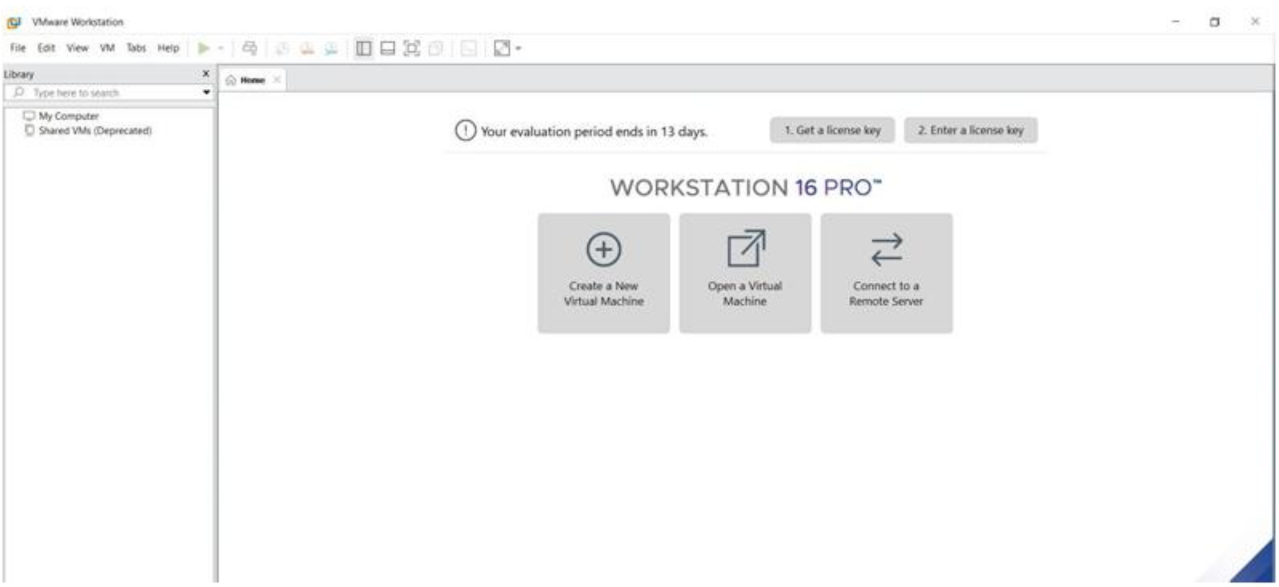
Algorithm

1. Install Oracle VirtualBox on the host operating system.
2. Download a Linux/Windows ISO image.
3. Open VirtualBox and create a new Virtual Machine.

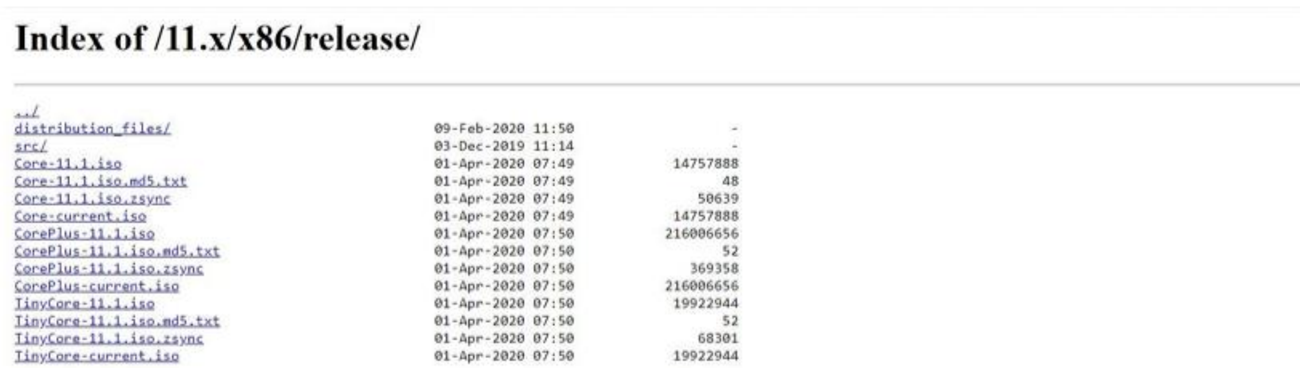
4. Select the VM name and operating system type.
5. Allocate 2 GB RAM to the VM.
6. Allocate 1 CPU to the VM.
7. Create a virtual hard disk with 15 GB storage.
8. Attach the downloaded ISO image to the VM.
9. Start the Virtual Machine.
10. Install the guest operating system.
11. Verify that the VM starts and works correctly.

Implementation Steps :

STEP 1:DOWLOAD VMWARE WORKSTATION AND INSTALLED AS TYPE 2 HYPERVISOR



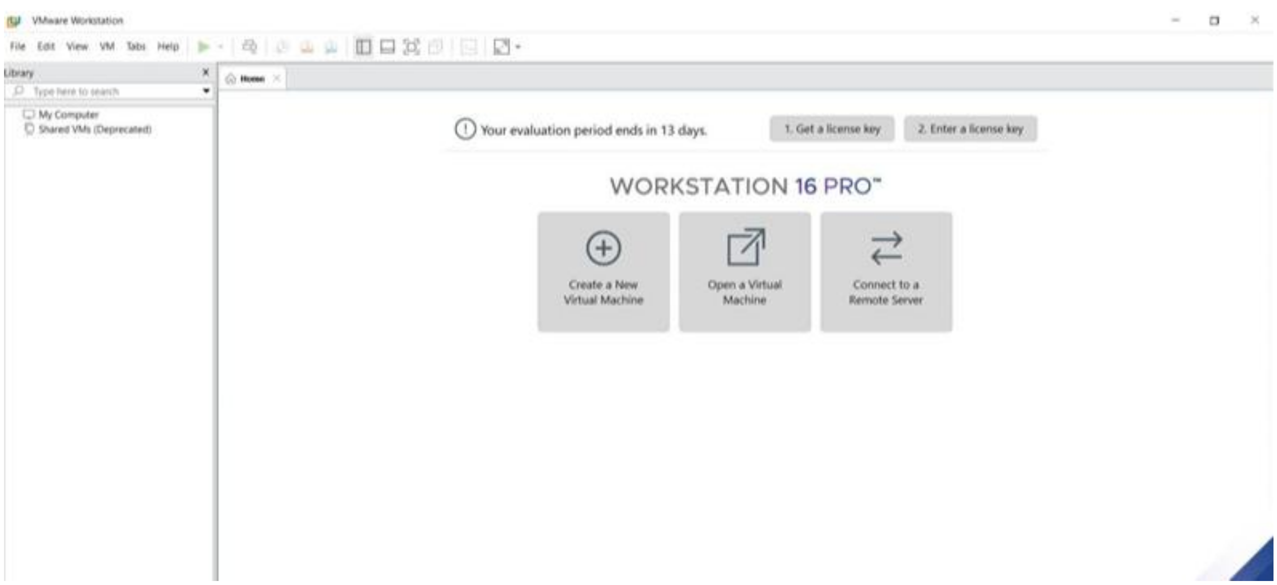
STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE



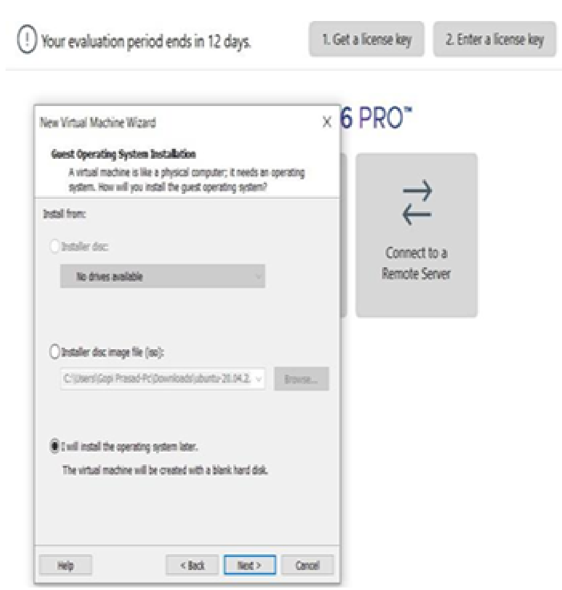
Index of /11.x/x86/release/

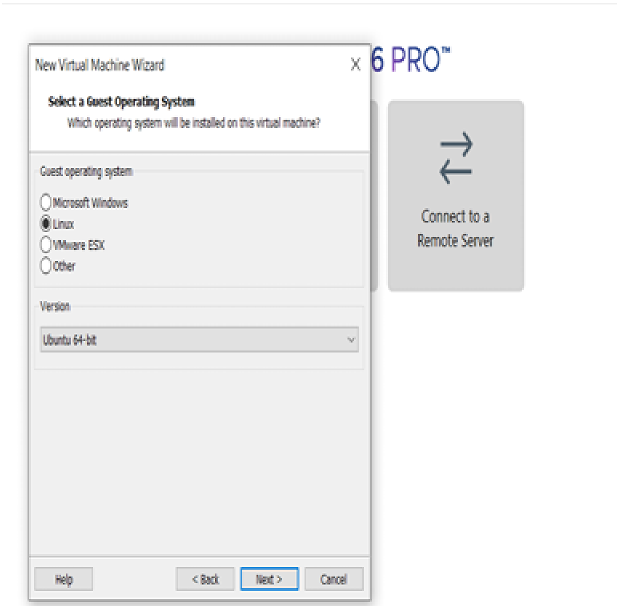
../		
distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.zsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

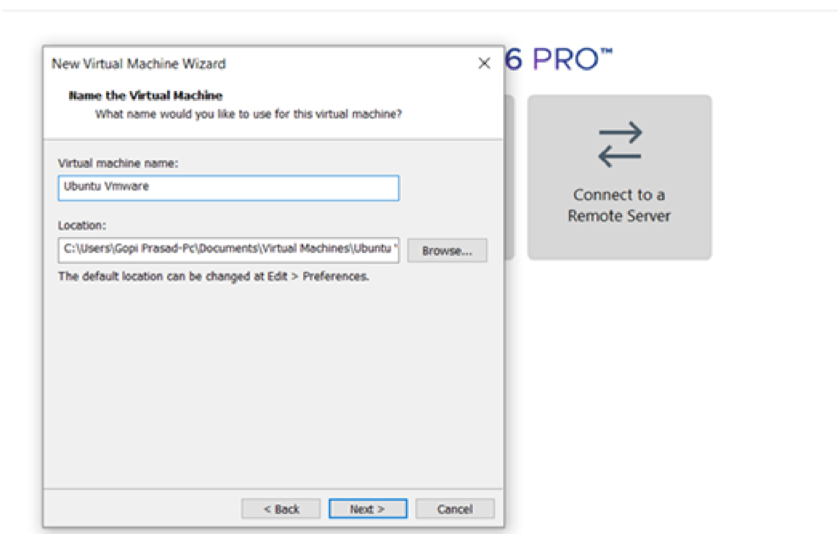
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



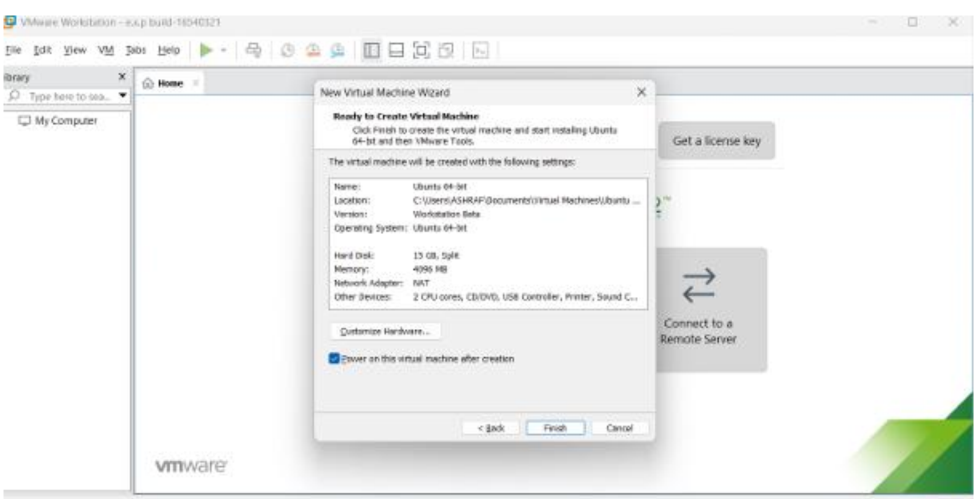
STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



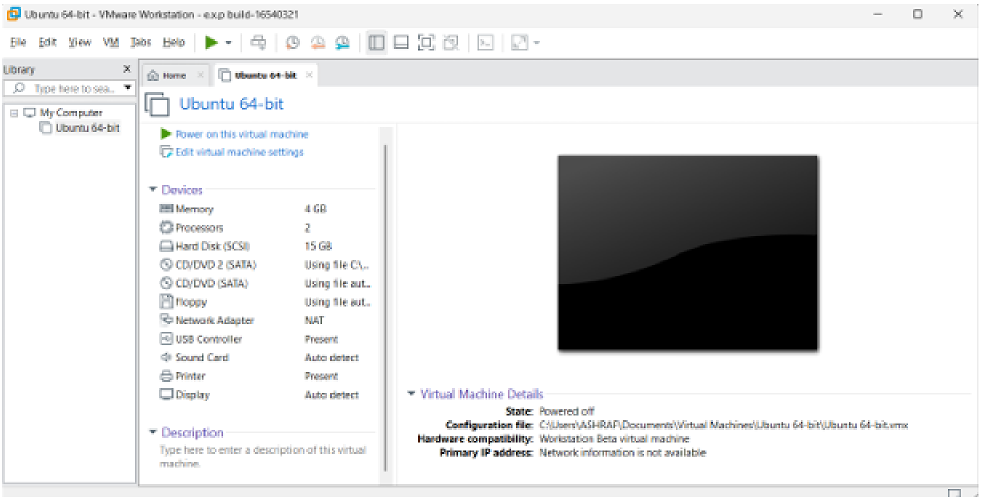




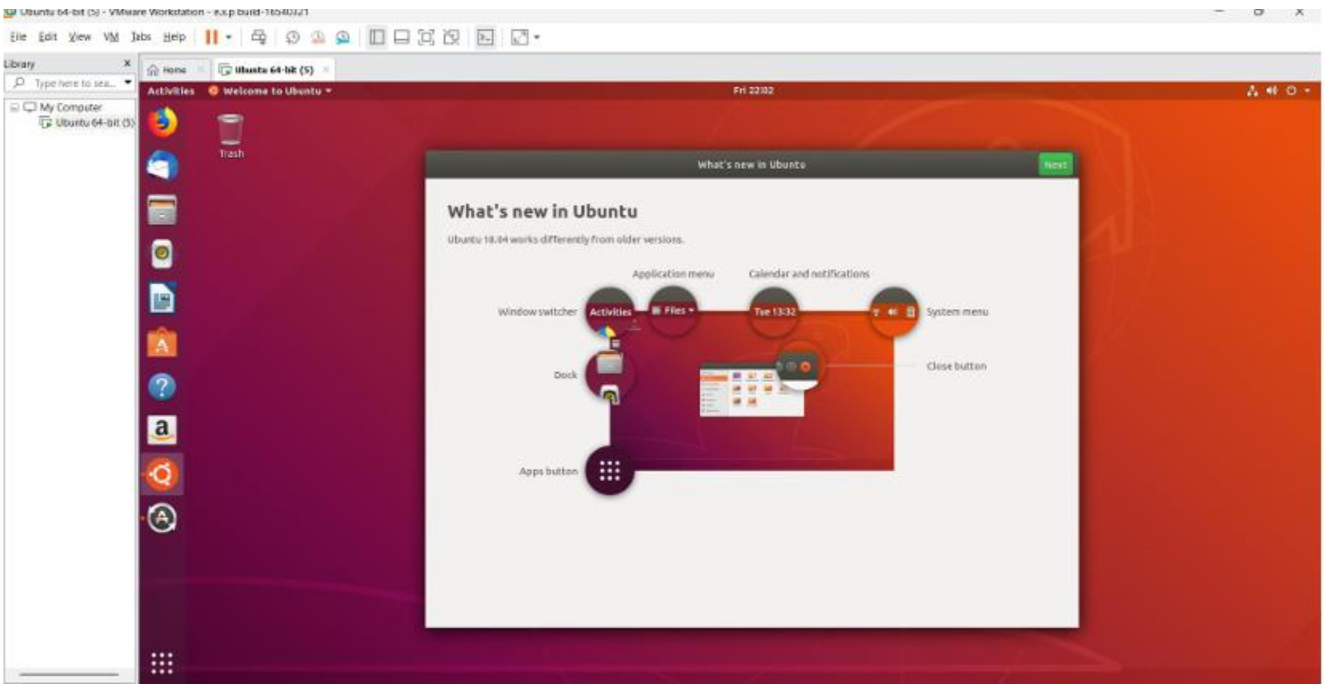
STEP 5: Set RAM = 2 GB and CPU = 1.



STEP 6: Set the hard disk size = 15 GB.



STEP 7: Start the VM and install Ubuntu



Result: The virtual machine was successfully created using VMware with 1 CPU, 2 GB RAM and 15 GB storage.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim

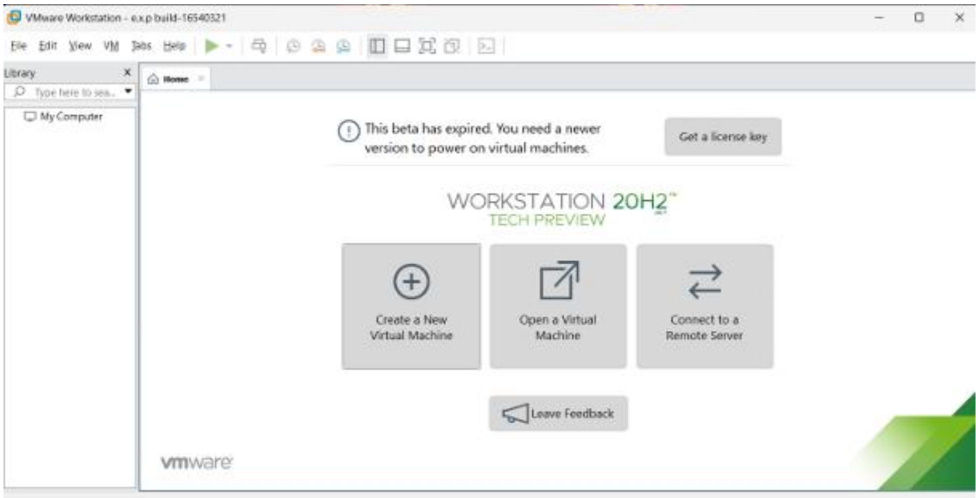
To demonstrate virtualization by creating a **clone of an existing Virtual Machine using VMware** and testing the cloned VM by loading the previous version.

Algorithm

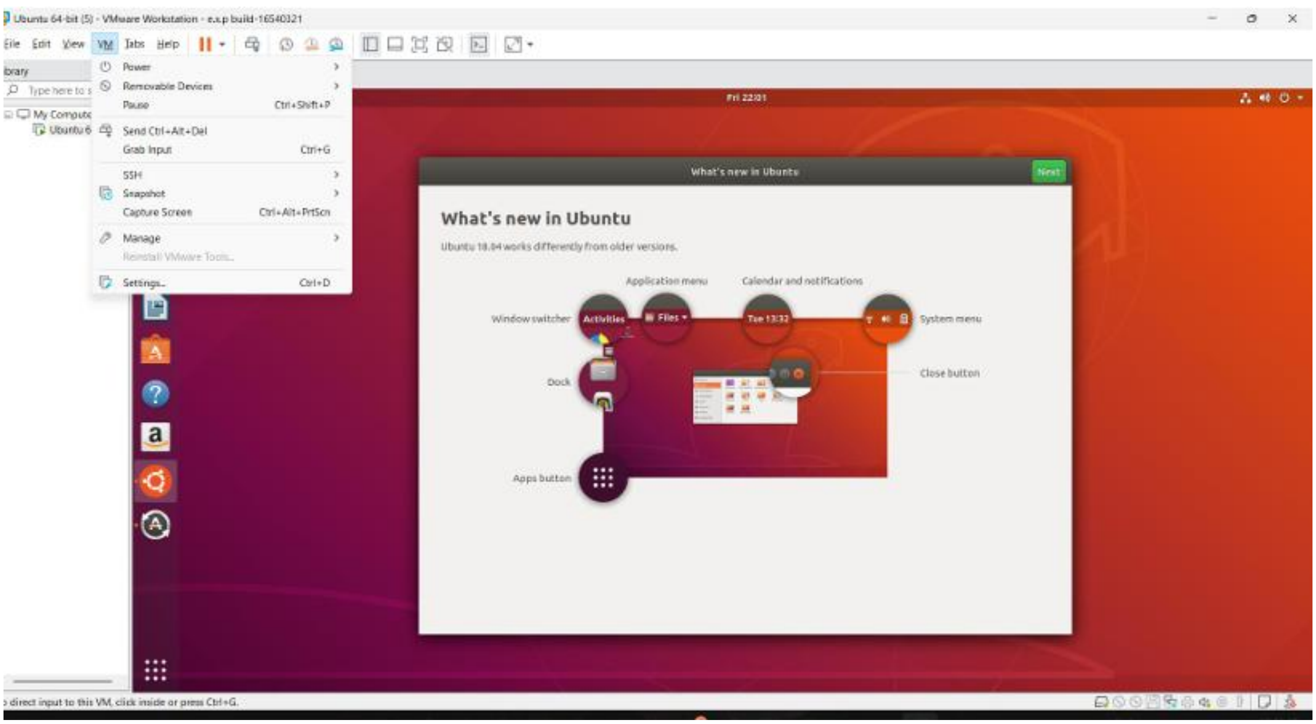
1. Open VMware Workstation.
2. Select the previously created VM.
3. Create a clone of the VM.
4. Select the appropriate cloning option.
5. Give a name and location for the cloned VM.
6. Complete the cloning process.
7. Start the cloned VM.
8. Verify the guest operating system and files.
9. Compare the cloned VM with the original VM.
10. Confirm successful cloning.

IMPLEMENTATION:

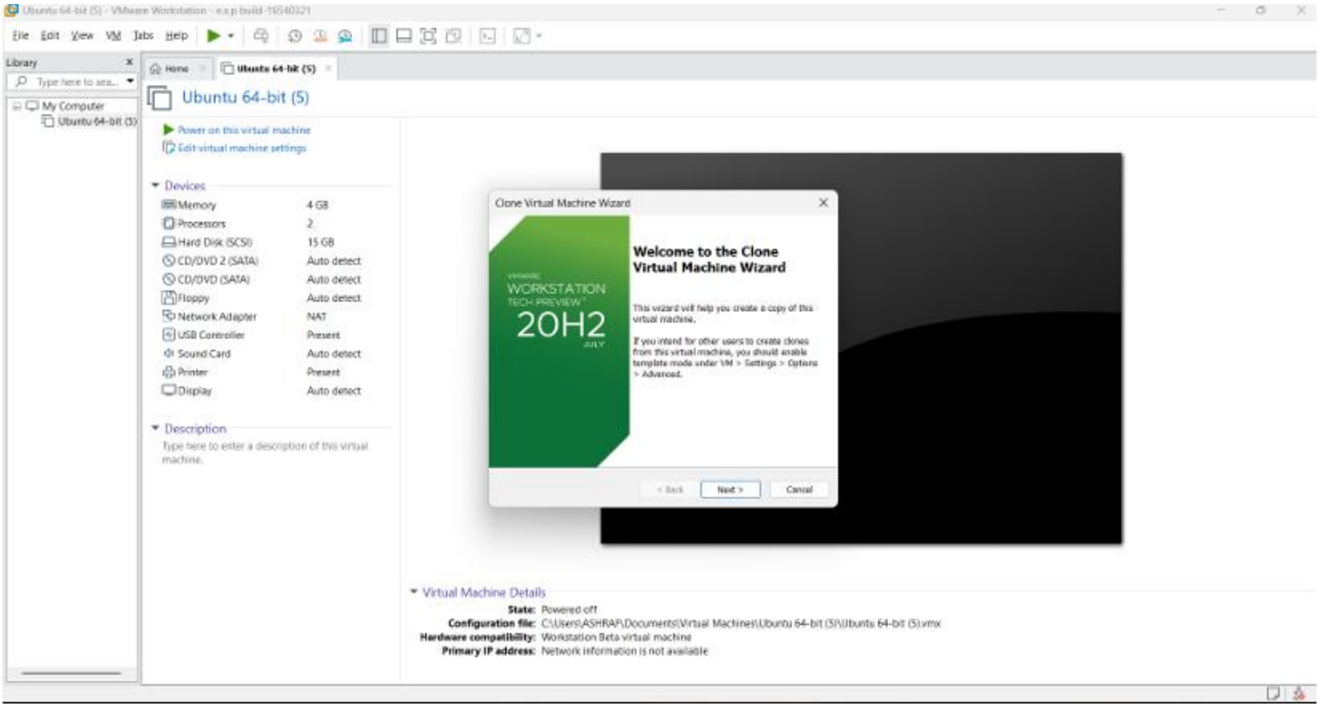
STEP 1: Open VMware Workstation, select the Ubuntu VM, and shut it down.



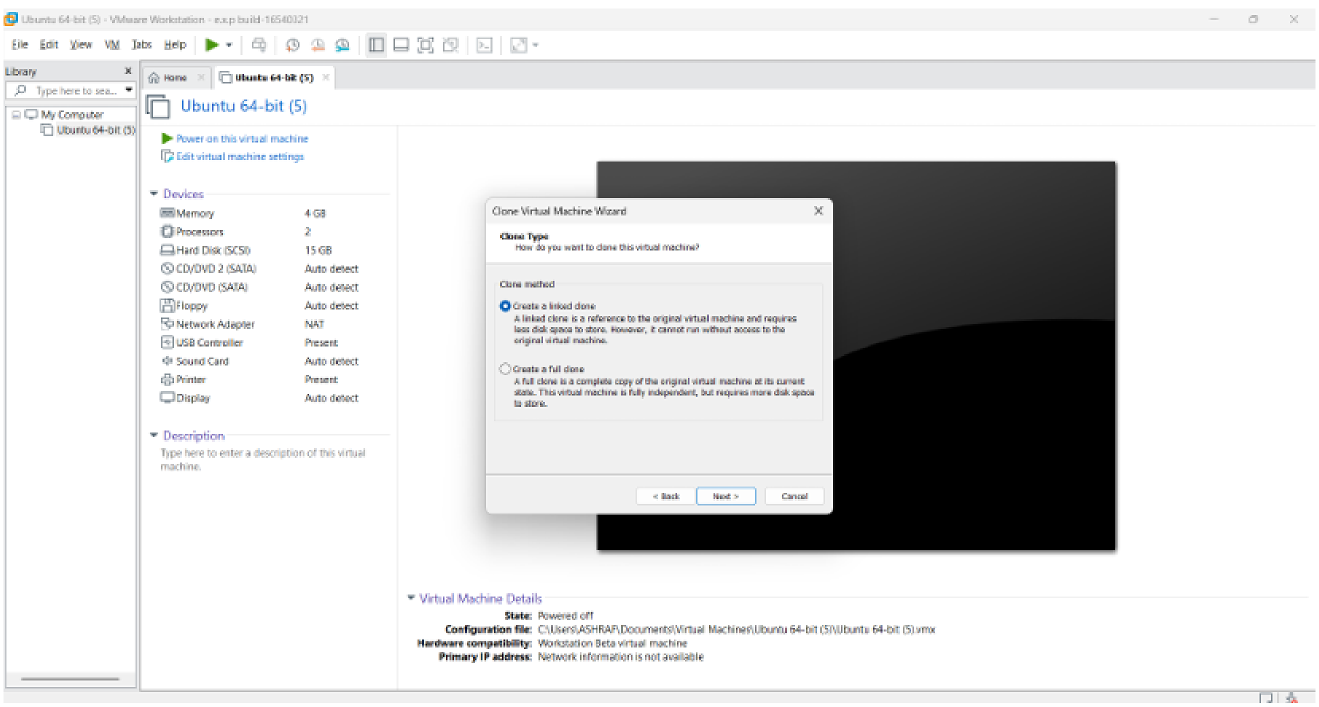
STEP 2: Select Manage → Clone to start the cloning process.



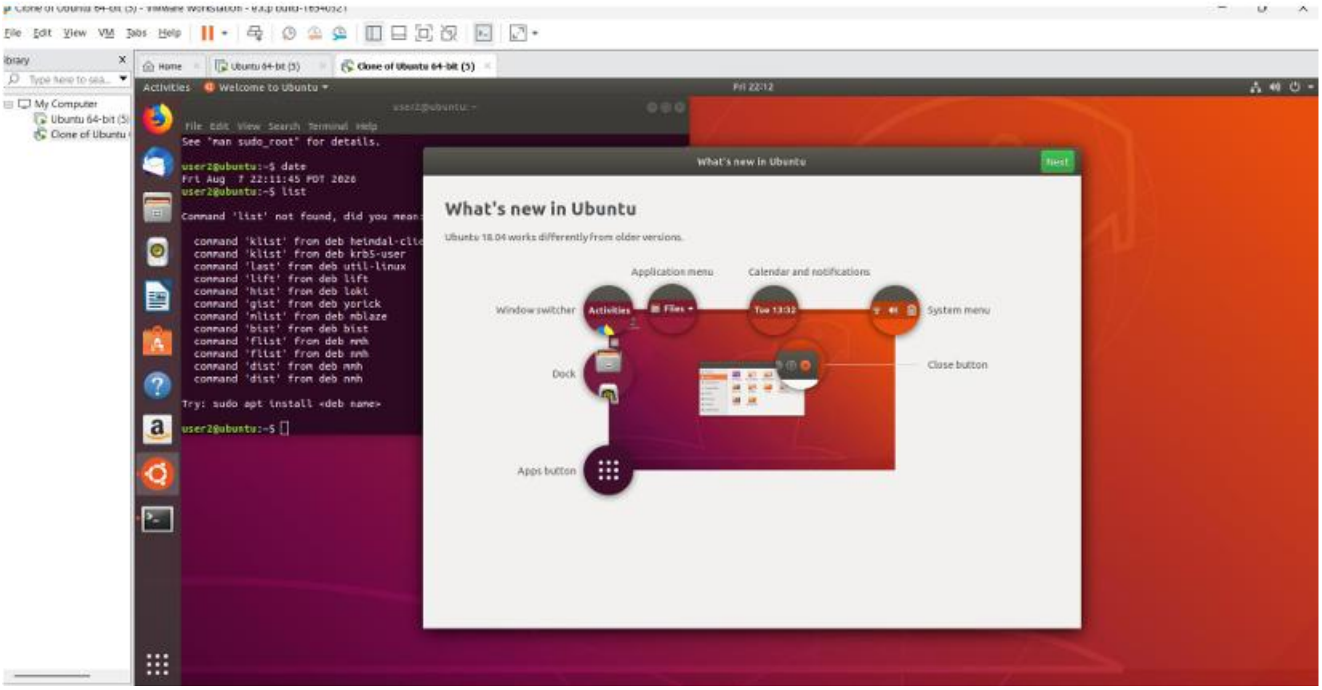
STEP 3: Select Full Clone and enter a name for the cloned VM.



STEP 4: Complete the cloning process and start the cloned VM.



STEP 5: Restore the previous snapshot, start the VM, and verify that the previous state is loaded.



Result :The VM was successfully cloned and tested, and its previous saved state was successfully restored